

Confounder-Robust Flood Detection from Street-Level Imagery via Phase-1 Hard Negative Mining

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Abstract

We introduce Phase-1 Hard Negative Mining (P1-HNM), a confounder-targeted resampling pipeline for street-level flood detection that exploits a partially trained intermediate checkpoint to recover visual confounder candidates that a fully converged model cannot provide. Existing flood screening systems report aggregate accuracy or ROC-AUC, which conceal category-level false positive rates and overstate discrimination on class-imbalanced datasets. Standard hard negative mining applied to fully converged models produces empty candidate sets because training convergence eliminates the decision-boundary uncertainty that mining requires. We fine-tune EfficientNetB0 and ResNet50 with a two-phase progressive transfer learning protocol on 4,099 deduplicated street-level images spanning eight non-flood categories including four water confounders, mine difficulty-ranked candidates from the Phase-1 checkpoint, apply $5\times$ augmentation, and retrain, ablating against extended-training and random-injection controls. We evaluate with PR-AUC as the primary metric and report per-category false positive rates with Clopper–Pearson 95 % binomial confidence intervals, validated with McNemar pairwise significance tests. EfficientNetB0 achieves PR-AUC= 0.9976 versus ResNet50’s 0.9614—a 3.6-point gap invisible in ROC-AUC (0.9985

vs. 0.9728)—and P1-HNM improves overall classification accuracy by 0.49 accuracy percentage points while difficulty-ranked selection outperforms both extended training and random augmentation alone, isolating the contribution of Phase-1 difficulty ranking.

Keywords: flood detection, hard negative mining, transfer learning, precision-recall AUC, visual confounders, deep learning, street-level imagery, EfficientNet, convolutional neural networks

1. Introduction

Floods are the most frequently occurring natural hazard worldwide and among the most destructive in terms of loss of life and economic damage. Satellite-based analysis covering 18 years of global data reveals that an average
5 of 58 million people were exposed to flooding each year between 2000 and 2018, with the exposed population increasing by 20–24 % over that period (Tellman et al., 2021). The economic costs of flood events continue to rise as urban development encroaches on flood plains and as extreme precipitation events become more frequent under climate change. Effective early warning
10 and rapid situational assessment are therefore critical infrastructure priorities.

Automated detection of flooding from crowdsourced street-level imagery offers a complementary situational awareness channel to traditional gauge networks and satellite revisit cycles. When citizens submit photographs via smartphone applications, social media platforms, or fixed traffic camera feeds,
15 the resulting image stream can provide hyperlocal evidence of active flooding that precedes official event reports by hours (Esparza et al., 2022). The first-pass screening problem is binary: given an image of unknown provenance

and unknown quality, does it depict active flooding? A screening system that operates at high recall on this binary problem enables downstream triage,
20 reducing the burden on human analysts who must validate alerts in real time.

The screening objective is recall-first. A missed flood image (false negative) means that an affected location is not escalated for human review; the consequence is delayed response and potentially preventable harm. A false positive triggers unnecessary resource allocation but carries a lower
25 immediate cost. This asymmetric cost structure makes flood recall—the fraction of actual flood images correctly identified—the primary operational metric. Performance depends strongly on the evaluation metric chosen, particularly given this recall-focused objective. Yet most published flood detection systems report ROC-AUC or overall accuracy, metrics that are
30 known to overstate performance on imbalanced binary classification problems (Davis and Goadrich, 2006): specificity is easy to achieve when negatives substantially outnumber positives, inflating ROC-AUC even when recall is poor. Precision-recall AUC (PR-AUC) is the more informative metric for this problem regime (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).
35 We demonstrate this gap empirically in Section 5.3: a 2.6-point ROC-AUC difference between EfficientNetB0 and ResNet50 corresponds to a 3.6-point PR-AUC difference and a difference of 56 missed floods in an 819-image validation set.

Visual confounders are the dominant failure mode. A flood de-
40 tection classifier trained on street-level images must distinguish flooded road surfaces from visually similar non-flood scenes. Swimming pools share an open, flat, reflective water surface, similar colour statistics under both overcast

and sunny illumination, and are embedded in urban and residential environments indistinguishable in scale from flooding. Rivers, particularly when
45 photographed at close range from bridges or embankments, produce images with similar surface texture, ripple patterns, and reflectivity to flooded streets. Park water features and ponds introduce the same water texture in urban green-space settings. Wet roads after rainfall present specular reflections on dark asphalt that partially mimic the low-albedo water signal of floods.
50 Under close-range photography, contextual cues that would disambiguate a pool from a flood at larger scale are absent: the horizon line, surrounding architecture, and ground plane are often cropped out or ambiguous. Standard cross-entropy trained models that exploit water texture as their primary flood cue will therefore misclassify all of these categories at rates that depend on
55 the overlap between flood and confounder visual distributions.

Per-category confounder analysis is absent from existing flood detection benchmarks. Existing benchmarks rarely provide fine-grained confounder category coverage, and published results report aggregate precision rather than per-category false positive rates with confidence intervals. Without
60 per-category statistics, a practitioner cannot determine which visual scenarios are driving false positives, nor can they target retraining efforts appropriately.

The hard negative mining signal problem. Standard offline hard negative mining (Viola and Jones, 2004; Shrivastava et al., 2016) addresses this by identifying the hardest-scoring non-flood images from a trained model and
65 injecting them into retraining to strengthen the decision boundary. However, this approach fails on fully converged two-phase fine-tuned models for a structural reason: by the end of Phase 2, the model assigns near-zero flood

probability to all training non-flood images regardless of their visual similarity to floods. The model has memorised the training distribution; mining from this checkpoint produces empty candidate lists. We address this by mining from the Phase-1 intermediate checkpoint, at which point the model has trained only the classification head and the top backbone layers. At this stage, the early ImageNet feature extractors are still frozen, leaving residual uncertainty on categories that share visual features with the positive class. The Phase-1 checkpoint retains genuine confusion on river and pool images that the Phase-2 checkpoint has resolved—this is the mining signal. This is a form of *offline* hard negative mining: candidates are identified in a single pre-training pass and injected into the dataset before retraining, rather than re-selected within each mini-batch.

Research gap. State-of-the-art ground-level flood screening systems achieve high aggregate performance—Lee and Joo (2025) report 97.67 % recall; Khan et al. (2023) report 96.43 % accuracy—yet none measure per-category confounder false positive rates with confidence intervals. Existing flood screening work does not evaluate FP rates broken down by confounder category, does not apply HNM from an intermediate checkpoint, and does not ablate the difficulty-ranking contribution against extended training and random data injection. The absence of ablation controls in prior HNM applications to flood data means it is unclear whether any observed improvements are attributable to the difficulty ranking itself or simply to the additional training epochs and data exposure. Our experimental design directly addresses this gap. We also measure per-category FP rates on a held-out validation set with Clopper–Pearson 95 % confidence intervals (Clopper and Pearson, 1934),

providing the statistical machinery to assess whether observed improvements are detectable at the available sample sizes.

95 **Contributions.** This paper makes the following contributions to the flood detection and remote sensing machine learning literature:

1. **Extended benchmark.** We combine the USF FloodingDataset (University of South Florida, 2023) with river images from the RIWA dataset (Wagner et al., 2023) and apply SHA-256 deduplication, producing
100 4,099 unique street-level images across 11 fine-grained categories with a stratified 80/20 train/val split (seed = 42). The benchmark includes fine-grained flood severity labels (Major, Moderate, Minor) enabling severity-stratified analysis.
2. **PR-AUC as primary metric.** We provide an empirical demonstration of a 3.6-point PR-AUC gap between EfficientNetB0 (0.9976) and
105 ResNet50 (0.9614) that is invisible in ROC-AUC (0.9985 vs. 0.9728), confirming Davis and Goadrich (2006)’s argument for PR-AUC on imbalanced screening problems and motivating its adoption as the standard primary metric for this class of problems.
- 110 3. **Phase-1 HNM pipeline.** A three-stage offline pipeline (Figure 4) that mines difficulty-ranked confounder candidates from the Phase-1 intermediate checkpoint, applies $5\times$ data augmentation, and retrains from the converged Phase-2 checkpoint. We release the pipeline implementation with the paper.
- 115 4. **Controlled ablation.** We isolate the difficulty-ranking contribution of P1-HNM by ablating against (a) extended training with no additional data and (b) random injection of category-matched non-flood images,

establishing the source of any improvement.

120 **5. Statistical validation.** Per-category false positive rates are reported with Clopper–Pearson 95 % exact binomial confidence intervals; model comparisons use McNemar’s pairwise test with Bonferroni correction; bootstrap confidence intervals are provided for multi-seed validation (pending additional seeds).

The remainder of the paper is organised as follows. Section 2 surveys
125 related work on flood detection, hard negative mining, and evaluation metrics. Section 3 describes the benchmark construction. Section 4 presents the training protocol, P1-HNM algorithm, and evaluation design. Section 5 reports experimental results. Section 6 interprets the findings and discusses limitations. Section 7 concludes.

130 **2. Related Work**

2.1. Ground-level flood detection

Ground-level flood detection from crowdsourced imagery has attracted increasing research attention as the prevalence of smartphone cameras and social media platforms provides a low-cost source of real-time visual evidence.
135 Esparza et al. (2022) examined data imbalance in crowdsourced flood reports submitted via a citizen science platform and demonstrated that class distribution directly affects model recall, motivating the use of weighted sampling strategies. Papadimos et al. (2023) benchmarked a range of multimedia flood detection approaches—including convolutional networks, graph
140 neural networks, and multimodal fusion—on a publicly available multimedia flood challenge dataset, finding that visual features alone achieve competitive

performance when training data is sufficient. Khan et al. (2023) proposed Flood-ResNet50, a ResNet50 variant optimised for deployment on resource-constrained edge hardware, reporting 96.43 % accuracy on their evaluation set and demonstrating inference throughput suitable for real-time operation. Lee and Joo (2025) introduced AlleyFloodNet, a dataset targeting economically marginalised alley environments—settings particularly vulnerable to surface flooding but underrepresented in existing benchmarks—and showed that a ConvNeXt-Large model fine-tuned on this data achieves 97.67 % flood recall. Alam et al. (2023) contributed the MEDIC benchmark for multi-task disaster image classification, covering floods, wildfires, hurricanes, and other events with multiple label types. A common limitation of all these systems is that none report per-category false positive rates for water-confounding non-flood categories; none distinguish river from pool from lake from wet-road confusion; and none report confidence intervals for small- N confounder categories. Our benchmark and evaluation protocol directly address this gap.

2.2. Aerial and satellite flood detection

A parallel literature addresses flood detection from aerial or satellite imagery. Rahnemoonfar et al. (2020) introduced FloodNet, a high-resolution post-Hurricane Harvey aerial imagery dataset with fine-grained semantic segmentation labels distinguishing flooded buildings, flooded roads, non-flooded buildings, and other scene elements. Tellman et al. (2021) developed a global framework using Landsat and MODIS satellite time series to estimate the number of people exposed to surface flooding, providing a macro-scale complement to event-level street-level detection. Aerial and satellite approaches benefit from consistent imaging geometry and large field of view, which resolve

many of the visual ambiguities that affect street-level classification—from altitude, a swimming pool and a flooded road are trivially distinguishable by context. Street-level screening therefore addresses a different, complementary
 170 problem: it provides hyperlocal, real-time evidence in the hours following a flood event before satellite revisit cycles produce updated imagery. Both approaches are needed in a complete early warning system.

2.3. Hard negative mining

Hard negative mining has a long history in detection pipelines. Viola and
 175 Jones (2004) demonstrated that bootstrapped negative mining—adding the hardest false positive windows from the current detector into the next round of training—dramatically reduces the false positive rate of the Viola-Jones face detector with minimal impact on detection rate. This insight was systematised as online hard example mining (OHEM) by Shrivastava et al. (2016): selecting
 180 the K highest-loss proposals within each mini-batch to focus gradient updates on informative examples. Lin et al. (2017) introduced focal loss as a continuous gradient-level analogue: down-weighting well-classified examples with a $(1 - p_t)^\gamma$ factor, automatically concentrating learning signal on hard cases without requiring explicit candidate selection. Hard negative mining methods differ
 185 along a key axis. *Online* methods (e.g., OHEM (Shrivastava et al., 2016) and focal loss (Lin et al., 2017)) re-select hard examples within each mini-batch during an in-progress training run. *Offline* methods identify candidates in a separate pass and inject them into the training dataset before the next training phase; they require a model that still produces informative difficulty scores on
 190 the candidate set. Our approach shares the goal of these methods but differs in three key respects. First, we use *offline batch* mining: candidates are identified

once per phase and injected into the training dataset, rather than re-selected per iteration. Second, we mine from the Phase-1 *intermediate* checkpoint rather than a fully converged model, because convergence eliminates the mining signal. Third, we operate at the *data level*—permanently adding new training examples—rather than at the gradient level. These differences are not incidental: the data-level injection is precisely what allows the difficulty-ranking contribution to be isolated from extended training and random injection in our ablation design.

2.4. *Progressive and catastrophic-forgetting-aware transfer learning*

Transfer learning from large-scale image classification models (*e.g.* ImageNet-pretrained ResNet50 (He et al., 2016) or EfficientNet (Tan and Le, 2019)) is standard practice for small-domain image classification problems. However, fine-tuning all layers simultaneously on a small target dataset risks catastrophic forgetting (Kirkpatrick et al., 2017): gradient updates for the new task overwrite previously learned representations, degrading generalisation. Progressive fine-tuning strategies mitigate this by gradually unfreezing layers from the task-specific head toward the backbone. Lyu et al. (2025) demonstrated that a staged ResNet50 protocol improves convergence stability and final accuracy on small datasets compared to full-network fine-tuning or head-only adaptation. Our two-phase protocol follows this design: Phase 1 trains the head and top backbone layers at a moderate learning rate, allowing the network to adapt its high-level features while preserving low-level texture representations; Phase 2 unfreezes deeper layers at a reduced learning rate, allowing gradual adaptation of mid-level features. The Phase-1 checkpoint thus captures an intermediate representation state that is informative for

mining but has not yet overfit the training distribution.

2.5. Evaluation metrics for imbalanced classification

The choice of evaluation metric is not a secondary concern for flood
220 screening but a primary one. Davis and Goadrich (2006) formally proved
that the ROC and precision-recall curves convey different information and
that the PR curve is more informative when the positive class prevalence is
low: ROC-AUC is dominated by the large number of true negatives (non-
flood images), inflating the apparent discrimination even when the classifier
225 systematically misclassifies the positive class. Saito and Rehmsmeier (2015)
provided empirical guidance for choosing between ROC-AUC and PR-AUC,
showing through extensive simulation that ROC-AUC misleads practitioners in
precisely the class-imbalanced regimes common in screening applications. The
critical difference is that PR-AUC penalises false negatives through the recall
230 axis and false positives through the precision axis, weighting them in a manner
directly relevant to the recall-first screening objective. For per-category false
positive rates on small validation sets, the normal approximation for binomial
confidence intervals fails badly (Clopper and Pearson, 1934); we use the exact
Clopper–Pearson interval, which provides guaranteed frequentist coverage
235 regardless of sample size. This is especially important for the swimming pool
category ($N=28$), where the upper bound of the 95 % CI for a zero observed
FP rate is 12.3 %.

3. Dataset

3.1. Source datasets

240 The flood images are drawn from the USF FloodingDataset (University of South Florida, 2023), a publicly available street-level benchmark providing binary flood/non-flood labels together with flood severity annotations at three levels: Major (severe inundation, water above knee level), Moderate (partial inundation, water on road surface), and Minor (early flooding, small puddles
245 visible). These severity classes represent operationally distinct scenarios for which recall requirements may differ: a Major flood image missed by the screener has higher operational cost than a Minor one. The non-flood categories in FloodingDataset include street scenes, animals, vehicles, buildings, plant life, and parks/walkways.

250 To supplement the river confounder category—which is the dominant source of false positives in our baseline models—we incorporate river images from the RIWA dataset (Wagner et al., 2023). RIWA is a pixel-wise binary river water segmentation collection of 1,128 labelled images captured with smartphones, drones, and DSLRs from diverse rivers, originally compiled to
255 benchmark convolutional network augmentation strategies for river surface segmentation. We use the image files without segmentation masks, treating all RIWA images as the ‘river’ non-flood subcategory. This enriches the river category to 399 images (323 train, 76 val), providing sufficient sample size to measure river FP rates with reasonable Clopper–Pearson confidence intervals.

260 3.2. Deduplication

Before splitting, we apply SHA-256 hash deduplication to the combined image pool to remove exact-duplicate files. This step is necessary because the source datasets may include identical images distributed under different filenames, and duplicate presence across train/val boundaries would artificially
265 inflate estimated generalisation performance. The deduplication step removed 55 exact-duplicate images, yielding 4,099 unique images. No perceptual similarity deduplication was applied; near-duplicate images (*e.g.* slightly different crops of the same scene) remain in the dataset.

3.3. Stratified split

270 Images are assigned to train (80 %) and validation (20 %) splits using stratified sampling by fine-grained category label with random seed = 42. Stratification ensures that the flood severity distribution (Major/Moderate/Minor) and the non-flood category distribution are preserved in both splits. Without stratification, random splits on an imbalanced dataset can produce train
275 or val sets that over-represent certain categories, biasing both the learned representations and the evaluation metrics. The resulting split contains 3,280 training images and 819 validation images. No test split is held out; threshold selection is performed on the validation set only, and the paper reports validation set performance throughout (Section 4.6).

280 3.4. Dataset statistics

Table 1 reports the image count per category and split. The flood class comprises three severity levels totalling 1,291 train and 322 val images. The non-flood class comprises eight fine-grained categories totalling 1,989 train

and 497 val images. Overall flood prevalence is 39.3% in train and 39.3% in
 285 val, confirming that stratification is effective and that neither split is class-
 balanced (*i.e.* 50/50 flood/non-flood). Class weights are therefore required to
 compensate for the 1:1.54 flood:non-flood ratio in training.

Table 1: Dataset statistics by category and split. Flood severity labels are Major, Moderate, and Minor. Non-flood categories marked † are water confounders. Note: swimming pool and plant have small validation counts ($N_{\text{val}} < 35$) that limit statistical power for per-category FP rate analysis.

Label	Category	Water?	Train	Val	Total
Flood	street_major	—	640	156	796
Flood	street_moderate	—	245	56	301
Flood	street_minor	—	406	110	516
<i>Flood total</i>			<i>1291</i>	<i>322</i>	<i>1613</i>
Non-flood	river†	Yes	323	76	399
Non-flood	swimming_pool†	Yes	70	28	98
Non-flood	park_walkway†	Partial	326	82	408
Non-flood	street_clear	No	475	105	580
Non-flood	animal	No	233	67	300
Non-flood	building	No	232	55	287
Non-flood	vehicle	No	189	50	239
Non-flood	plant	No	141	34	175
<i>Non-flood total</i>			<i>1989</i>	<i>497</i>	<i>2486</i>
Total			3280	819	4099

The smallest validation category is swimming pool ($N_{\text{val}} = 28$). The upper

bound of the Clopper–Pearson 95 % CI for an observed FP rate of 0 % on this
category is 12.3 % (Section Appendix B). This means that a null result on
swimming pool cannot be interpreted as evidence of confounder robustness.
Similarly, plant ($N_{\text{val}} = 34$, upper CI bound 10.3 %) and animal ($N_{\text{val}} = 67$,
upper CI bound 5.4 %) have validation counts too small for high-confidence
conclusions. Expanding these categories is an important priority for future
work.

4. Methods

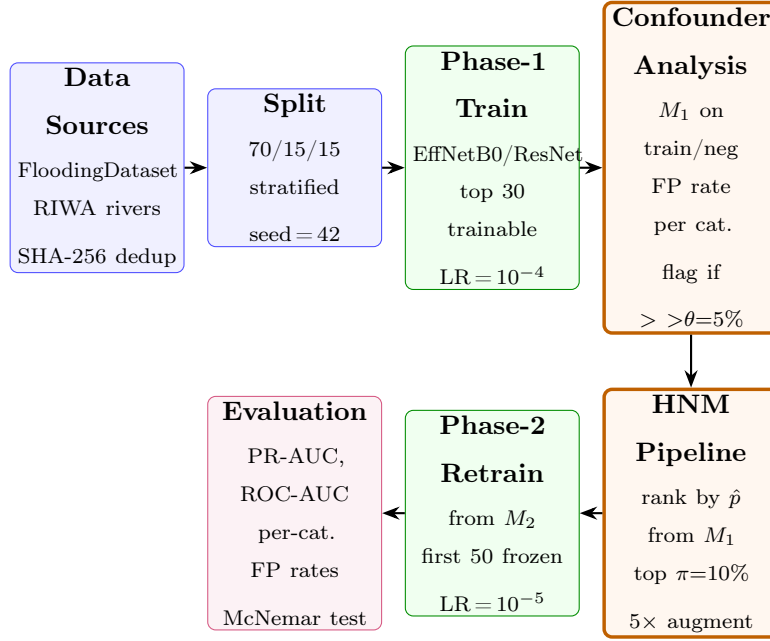


Figure 1: End-to-end methodology pipeline. Blue boxes: data preparation stages; green boxes: training stages; orange boxes (bold border): Phase-1 HNM novel contribution (Section 4.4); purple box: evaluation. Data flows left-to-right in the top row; the HNM pipeline mines difficulty-ranked candidates from M_1 and retrains from M_2 .

4.1. Architectures

We evaluate two ImageNet-pretrained convolutional architectures to assess whether the P1-HNM pipeline is architecture-agnostic or specific to particular
300 feature spaces.

EfficientNetB0 (Tan and Le, 2019). EfficientNet scales network depth, width, and input resolution simultaneously via a compound scaling coefficient, producing better accuracy per parameter than architectures scaled on a single dimension. EfficientNetB0 has approximately 5.3 M parameters. Under Keras
305 3, the `include_preprocessing` argument has been removed; the built-in rescaling layer inside the model graph handles pixel normalisation, so the model receives raw $[0, 255]$ pixel values and performs its own $\text{scale} = \frac{1}{255}$ rescaling internally. This is a critical implementation detail: passing pre-normalised inputs to EfficientNetB0 in Keras 3 produces degraded performance.

310 **ResNet50** (He et al., 2016). ResNet50 uses shortcut connections to train networks of 50 residual layers with approximately 25.6 M parameters. Input is preprocessed by channel-mean subtraction using the ImageNet channel means $(\mu_R, \mu_G, \mu_B) = (103.94, 116.78, 123.68)$ via `tf.keras.applications.resnet50.preprocess_input`. Mixing the preprocessing conventions between architectures (*i.e.* applying
315 ResNet preprocessing to EfficientNetB0 or vice versa) is a documented source of performance degradation in flood-screening code and was verified at runtime in our implementation.

Classification head. Both architectures use a shared binary classification head appended after the backbone:

$$\text{GAP}(\mathbf{x}) \xrightarrow{\text{Drop}(0.2)} \text{Dense}(256, \text{ReLU}) \xrightarrow{\text{BN}} \xrightarrow{\text{Drop}(0.3)} \text{Dense}(1, \sigma) \quad (1)$$

320 where GAP is global average pooling, BN is batch normalisation, $\text{Drop}(p)$ is

dropout with drop probability p , and σ is the sigmoid activation. The output is a scalar $\hat{p} \in [0, 1]$ representing $P(\text{non-flood})$. The flood probability used for all evaluation is $1 - \hat{p}$. Figure 2 illustrates the full architecture.

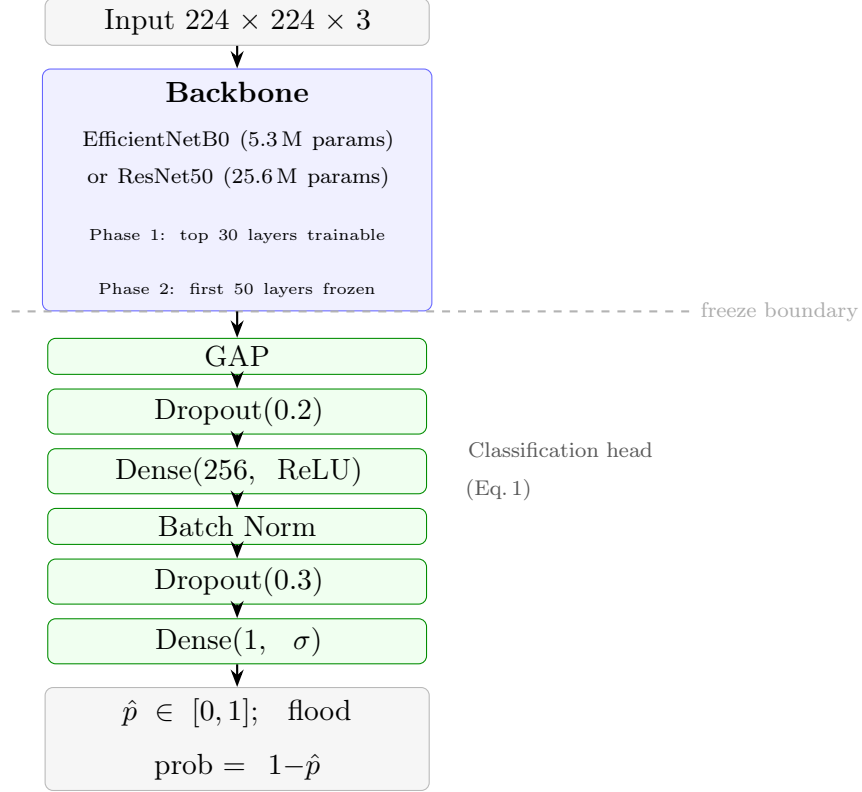


Figure 2: Model architecture. The ImageNet backbone feeds a shared binary classification head (Eq. 1). The dashed line marks the freeze boundary between Phase 1 (top 30 backbone layers trainable) and Phase 2 (first 50 backbone layers frozen). GAP = global average pooling. The output $\hat{p}$ is the non-flood probability; flood probability used in all evaluation = $1 - \hat{p}$.

4.2. Two-phase progressive fine-tuning

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Two-phase progressive fine-tuning mitigates catastrophic forgetting (Kirkpatrick et al., 2017) and has been shown to improve convergence stability on

small datasets compared to full-network fine-tuning (Lyu et al., 2025). Our two-phase protocol is defined as follows.

Phase 1. Freeze all backbone layers except the last 30 (the layers closest
330 to the classification head, counting from the output). Train the classification head and these top 30 backbone layers at $\text{LR}_1 = 1 \times 10^{-4}$ for up to $T_1 = 15$ epochs. The rationale is that the high-level semantic features encoded in the top layers are the most task-specific and require the most adaptation, whereas the low-level texture features in the early layers are general-purpose
335 and should be preserved.

Phase 2. Freeze the first 50 backbone layers (the earliest feature extractors from the input). Fine-tune all remaining layers at $\text{LR}_2 = 1 \times 10^{-5}$ for up to $T_2 = 20$ epochs starting from the best Phase-1 checkpoint. The reduced learning rate prevents large gradient steps from disrupting the representation
340 quality already achieved in Phase 1.

Both phases use: (a) early stopping with patience = 7 epochs on validation loss, saving the model at the epoch with lowest validation loss; (b) `ReduceLROnPlateau` with patience = 3 (Phase 1) and patience = 4 (Phase 2), halving the learning rate when validation loss fails to improve; (c) balanced
345 class weights computed per run via `sklearn.utils.class_weight.compute_class_weight("balanced", ...)` on the training label distribution, producing weights inversely proportional to class frequency.

The full set of hyperparameters is listed in Section Appendix A. Figure 3 summarises the two-phase protocol and shows how the P1-HNM pipeline uses
350 the two checkpoints.

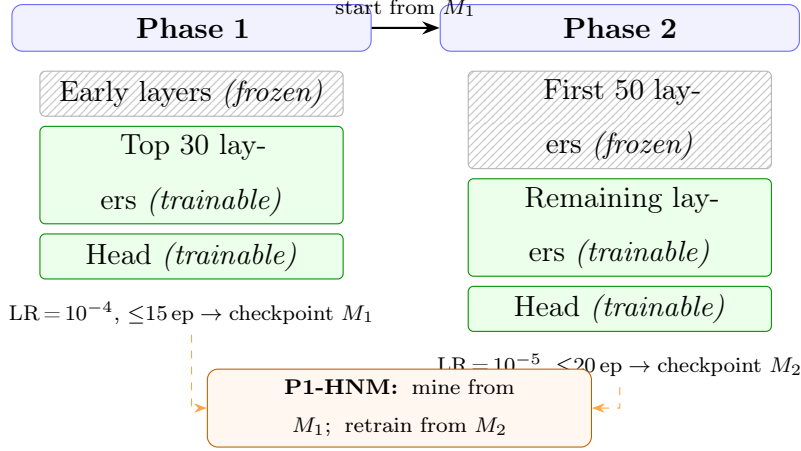


Figure 3: Two-phase progressive fine-tuning schematic. Hatched boxes = frozen layers; green boxes = trainable layers. Phase 1 trains the head and top 30 backbone layers at $\text{LR} = 10^{-4}$ for up to 15 epochs (checkpoint M_1). Phase 2 freezes the first 50 backbone layers and fine-tunes the remainder at $\text{LR} = 10^{-5}$ for up to 20 epochs (checkpoint M_2). The orange box shows how P1-HNM uses both checkpoints.

4.3. Loss functions

Binary cross-entropy (BCE). Standard sigmoid cross-entropy:

$$\mathcal{L}_{\text{BCE}} = -w_y [y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

where $y \in \{0, 1\}$ is the true label ($1 = \text{flood}$) and w_y is the class weight for class y . Class weights provide a global frequency correction for the flood:non-flood
 355 $\approx 1 : 1.54$ imbalance in training.

Focal loss (Lin et al., 2017). Focal loss addresses two problems simultaneously: class imbalance (via the class weight w_y) and hard example weighting (via a modulating factor that down-weights easy, well-classified examples within each mini-batch). We evaluate it as an alternative to BCE to test
 360 whether local hard example weighting provides additional benefit on top of

global frequency correction.

$$\mathcal{L}_{\text{focal}} = -w_y(1 - p_t)^\gamma [y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (3)$$

where $p_t = \hat{p}$ if $y = 1$ and $p_t = 1 - \hat{p}$ if $y = 0$, and $\gamma = 2.0$. The factor $(1 - p_t)^\gamma$ down-weights well-classified examples within each mini-batch, concentrating gradient signal on hard cases. We combine focal loss with balanced class weights, so class weighting provides global frequency correction while focal loss provides local within-batch difficulty correction. These are distinct orthogonal interventions: a class-weighted focal loss model up-weights the flood class globally *and* down-weights easy flood examples locally.

4.4. Phase-1 hard negative mining

The P1-HNM pipeline is illustrated in Figure 4 and proceeds in three stages.

4.4.1. Stage 1: Confounder analysis from the Phase-1 checkpoint

The Phase-1 checkpoint M_1 is run on all `train/non_flood/` images. For each image, the flood probability is computed as $\hat{p} = 1 - \sigma(\text{logit})$, where σ is the sigmoid activation and the logit corresponds to the non-flood class due to alphabetical class ordering in Keras’s `flow_from_directory` (class index 0 = flood, 1 = non-flood). The per-category FP rate is computed at threshold $\tau = 0.5$, and any category with FP rate exceeding $\theta = 0.05$ is flagged.

The central motivation for using the Phase-1 checkpoint as the mining oracle is that training convergence destroys the mining signal. By the end of Phase 2, the fully converged model assigns near-zero flood probability to all training non-flood images regardless of visual similarity: EfficientNetB0

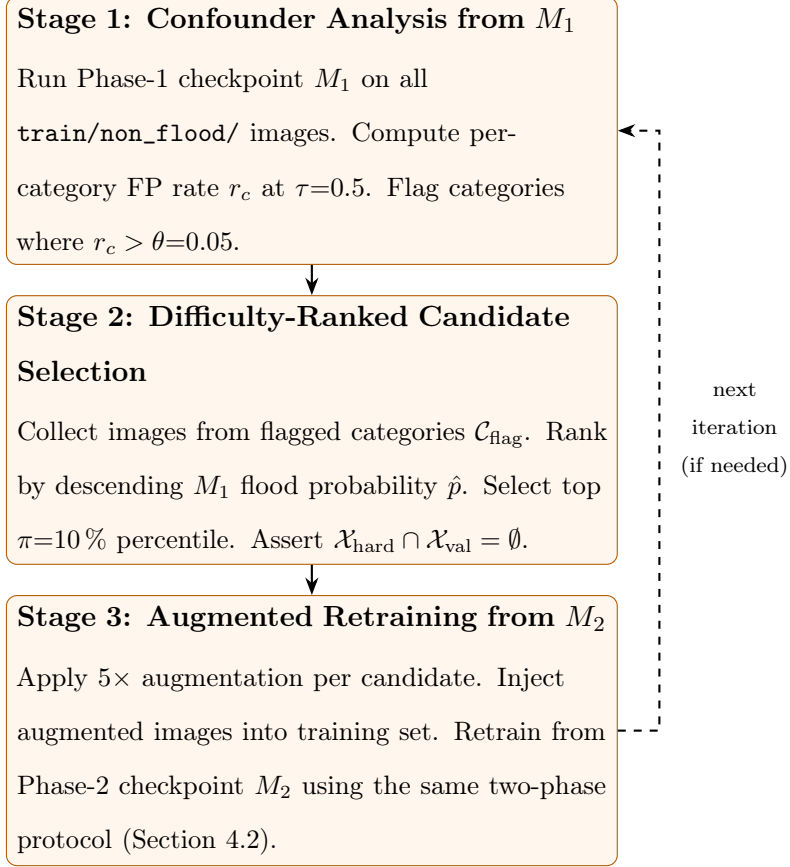


Figure 4: P1-HNM three-stage pipeline. Stage 1 identifies high-FP confounder categories using the Phase-1 checkpoint M_1 , which retains genuine uncertainty on visual confounders because only the head and top 30 backbone layers have been trained. Stage 2 selects the hardest candidates by difficulty ranking. Stage 3 retrains from the converged Phase-2 checkpoint M_2 with augmented hard negatives. The dashed feedback arrow indicates multi-iteration deployment.

Phase-2 achieves a training-set river FP rate of 0.31 %, leaving fewer than two candidates above the 0.5 threshold across all 323 training river images. The
 385 Phase-1 checkpoint has trained only the head and the top 30 backbone layers; the early ImageNet feature extractors (~ 70 % of the backbone) remain frozen. These frozen layers produce feature representations calibrated on ImageNet rather than on flood data, retaining genuine uncertainty on river and pool images whose textures resemble flood water in the ImageNet feature space.
 390 This residual uncertainty provides the difficulty signal that Stage 2 exploits.

4.4.2. Stage 2: Difficulty-ranked candidate selection

From the flagged categories, all images are collected and ranked by descending Phase-1 flood probability $\hat{p}$. The top $\pi = 10$ % are selected as hard negative candidates. Percentile-based selection is invariant to probability
 395 calibration—it selects the images on which the Phase-1 model is most confused, not merely those that exceeded a binary threshold. This is important because the calibration of the Phase-1 model is not optimised; using an absolute probability threshold would produce inconsistent candidate set sizes across architectures and loss functions.

400 A partition-safety assertion is enforced before proceeding: every candidate in $\mathcal{X}_{\text{hard}}$ is confirmed to be absent from the validation set by checking file paths. This ensures that the hard negative injection does not constitute test-set leakage and that the resulting model’s validation performance reflects genuine generalisation.

4.4.3. Stage 3: Augmented retraining from the Phase-2 checkpoint

Each selected hard negative image receives $5\times$ data augmentation: random rotation $\pm 30^\circ$, random zoom $\pm 30\%$, random brightness scaling by a factor in $[0.6, 1.4]$, and random horizontal flip. The augmented images are injected into the training set alongside the original candidates. This $5\times$ factor increases the effective weight of the hard negatives in the training distribution without requiring loss reweighting.

The augmented training set is used to retrain from the Phase-2 checkpoint M_2 using the same two-phase protocol described in Section 4.2. Starting from M_2 rather than from scratch ensures that the HNM model benefits from all the representation learning performed in the original two-phase training; the Phase-2 checkpoint is the strong converged baseline and serves as a warm start for the confounder-targeted refinement.

4.5. Ablation controls

Three ablation conditions are defined to isolate the contribution of difficulty-ranked selection from confounding factors. Table 2 summarises the conditions.

Table 2: Ablation conditions isolating the contributions of additional training and difficulty ranking. All conditions start from the Phase-2 checkpoint M_2 ; they differ only in what data is added to the training set.

Condition	Description	Controls for
HNM (percentile)	Top 10 % hardest negatives by M_1 probability	— (target condition)
Extended training	Same epoch budget; no additional data	Additional training epochs
Random injection	Same count; randomly sampled from flagged categories	Category exposure vs. diff

The **extended training** control answers the question: does additional training from the Phase-2 checkpoint improve performance without any new data? If HNM outperforms extended training, the improvement cannot be attributed solely to additional compute.

425 The **random injection** control answers a more specific question: is difficulty-ranked selection necessary, or is any non-flood data from the flagged categories sufficient? If HNM outperforms random injection, the improvement is attributable to the difficulty ranking itself, not merely to additional exposure to the confounder categories. Together, the three-way comparison provides a
430 hierarchical isolation of the P1-HNM contribution.

4.6. Evaluation protocol

Primary metric: PR-AUC. We motivate the use of PR-AUC over ROC-AUC in Section 5.3. PR-AUC is computed using `sklearn.metrics.precision_recall_curve` and `sklearn.metrics.auc` on the validation set.

435 **Threshold-level metrics.** At $\tau = 0.5$ we report: flood recall ($TP/(TP+FN)$), flood precision ($TP/(TP+FP)$), flood F1, accuracy, false negative count (FN), and false positive count (FP). The threshold is fixed at 0.5 for all conditions; no threshold optimisation on the validation set is performed.

Per-category FP rates. For each non-flood category c with n_c validation
440 images, the observed FP count k_c (non-flood images predicted as flood) is recorded and the FP rate $r_c = k_c/n_c$ is reported with a Clopper–Pearson 95 % binomial CI:

$$[B(\alpha/2; k_c, n_c - k_c + 1), B(1 - \alpha/2; k_c + 1, n_c - k_c)] \quad (4)$$

where $B(q; a, b)$ denotes the q -quantile of the $\text{Beta}(a, b)$ distribution and

$\alpha = 0.05$.

445 **Statistical comparisons.** Pairwise model comparisons use McNemar’s test (McNemar, 1947) on paired validation predictions with Bonferroni correction for the number of pairwise comparisons. For per-category FP comparisons across conditions, Fisher’s exact test is used due to small cell counts.

5. Experiments and Results

450 In this section we present the baseline training dynamics (Section 5.1), overall discrimination metrics (Section 5.2), the PR-AUC/ROC-AUC discrepancy (Section 5.3), qualitative error analysis (Section 5.4), per-category confounder analysis (Section 5.5), and P1-HNM ablation results (Section 5.6).

5.1. Baseline training dynamics

455 Figure 5 shows training and validation curves for the EfficientNetB0 BCE baseline over 20 epochs (the last Phase-2 epoch; Phase-1 runs for a maximum of 15 epochs internally). The Phase-1 to Phase-2 boundary is visible around epoch 7–8 as a transient spike in training loss: when the frozen backbone layers are unfrozen and the learning rate is reduced from 1×10^{-4} to 1×10^{-5} ,
460 the model briefly destabilises as previously frozen layer weights are exposed to gradients for the first time, before re-converging. This spike is expected under progressive fine-tuning and does not indicate training instability.

After the spike, training loss decreases smoothly and validation loss stabilises around 0.045–0.050. The train-validation loss gap of approximately
465 0.02 at convergence indicates mild overfitting, consistent with a 3,280-image training set. The validation AUC (ROC) reaches 0.9980 at the best epoch and stays above 0.9960 for all final epochs, indicating stable generalisation. The

Keras `Recall` metric tracked during training measures non-flood recall (class index 1, alphabetically first in binary classification), not flood recall; flood
 470 recall is computed from the confusion matrix in post-training evaluation.

ResNet50 exhibits different dynamics: it achieves a stable Phase-2 plateau earlier and terminates via early stopping after fewer total epochs. Its validation AUC reaches 0.9986 at the best epoch—slightly above EfficientNetB0—yet its per-threshold performance (*e.g.* flood recall at $\tau = 0.5$) is substantially
 475 worse, illustrating the disconnect between ROC-AUC and practical operating performance.

5.2. Baseline performance at $\tau = 0.5$

Table 3 reports validation set performance for all four baseline conditions at the fixed decision threshold $\tau = 0.5$. We discuss flood precision
 480 (TP/(TP+FP)) and flood recall (TP/(TP+FN)) over 322 flood validation images and 497 non-flood validation images.

EfficientNetB0 BCE. EfficientNetB0 BCE achieves 97.8% flood recall with only 7 false negatives and 7 false positives. The nearly symmetric error pattern (7 FN, 7 FP) indicates that the model has identified a well-calibrated
 485 decision boundary near the ambiguous boundary between flood and near-flood-like non-flood categories. PR-AUC of 0.9976 confirms that precision remains high across the entire recall range.

EfficientNetB0 Focal. Focal loss ($\gamma = 2.0$) modestly improves EfficientNetB0: flood recall increases from 97.8% to 98.5% (7 FN to 5 FN) and FP
 490 decreases from 7 to 5. PR-AUC improves marginally from 0.9976 to 0.9977. The consistent improvement across both FN and FP suggests that focal loss helps EfficientNetB0 find a slightly better decision boundary, consistent

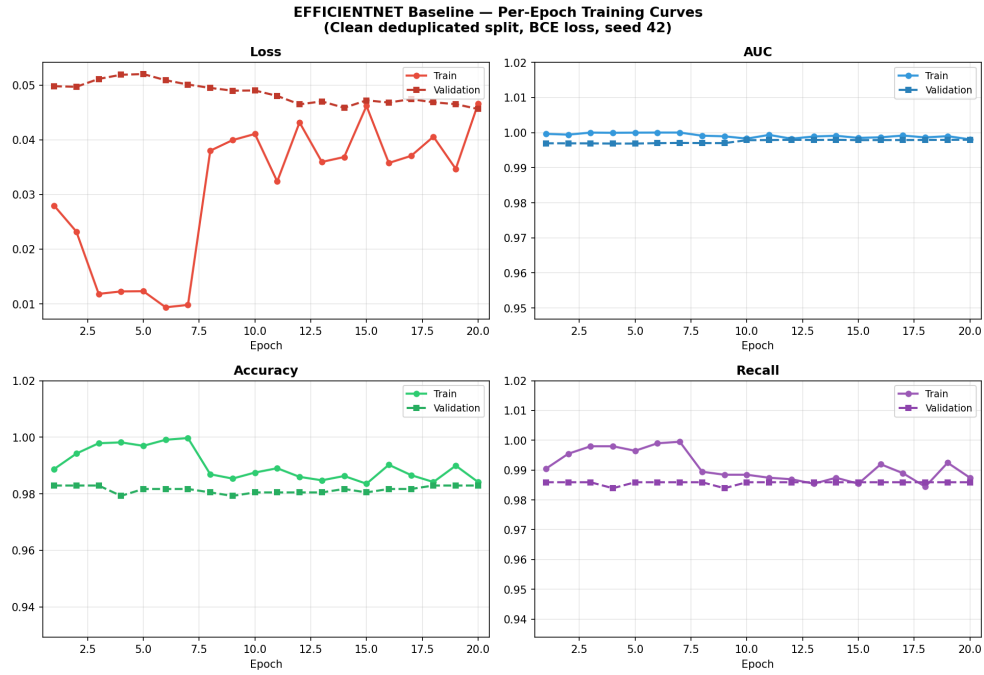


Figure 5: EfficientNetB0 BCE baseline training curves (seed 42, 80/20 clean split). Solid lines = training; dashed = validation. The training loss spike at epoch ~7–8 marks the Phase-1 to Phase-2 transition when frozen backbone layers are unfrozen. All metrics are Keras monitored values; recall here is non-flood class recall (class index 1). Best epoch (lowest val loss) is epoch 20.

with the compound-scaled architecture’s capacity to benefit from gradient reweighting.

495 **ResNet50 BCE.** ResNet50 BCE produces a highly asymmetric error pattern: 63 FN and 17 FP. The 19.6 % flood miss rate (63/322) is operationally unacceptable for a first-pass screening system. Despite this, ResNet50 achieves *higher* overall accuracy (98.5 % vs. EfficientNetB0’s 98.3 %), illustrating the accuracy paradox discussed in Section 5.3: the larger non-flood class (497
500 images) dominates the accuracy denominator, masking the systematic flood miss rate. PR-AUC of 0.9614 correctly captures this failure.

ResNet50 Focal. Focal loss substantially *degrades* ResNet50: recall drops from 80.4 % to 71.1 % (63 FN to 93 FN), while FP decreases from 17 to 10. The net effect is a model with worse flood recall but better non-flood
505 precision—the opposite of the desired recall-first objective. PR-AUC decreases from 0.9614 to 0.9600. This divergence between focal loss effects on the two architectures is discussed further in Section 6.2.

Table 3: Baseline validation-set performance at $\tau=0.5$ (seed 42; 819 val images: 322 flood, 497 non-flood). Bold marks best per-column value. FN = missed floods; FP = spurious alerts. Flood recall = $TP/(TP+FN)$; Flood Prec. = $TP/(TP+FP)$.

Model	Loss	Acc (%)	Rec (%)	Prec.	F1	FN	FP	PR-AUC	ROC-AUC
EfficientNetB0	BCE	98.3	97.8	0.978	0.978	7	7	0.9976	0.9985
EfficientNetB0	Focal	98.8	98.5	0.985	0.985	5	5	0.9977	0.9986
ResNet50	BCE	98.5	80.4	0.938	0.866	63	17	0.9614	0.9728
ResNet50	Focal	87.4	71.1	0.958	0.816	93	10	0.9600	0.9721

Figure 6 shows confusion matrices for all four conditions. The visual

contrast between EfficientNetB0 (near-symmetric errors, blue upper-left cell
 510 dominant) and ResNet50 (heavily asymmetric, many blue lower-left FN cells)
 makes the operational difference concrete.

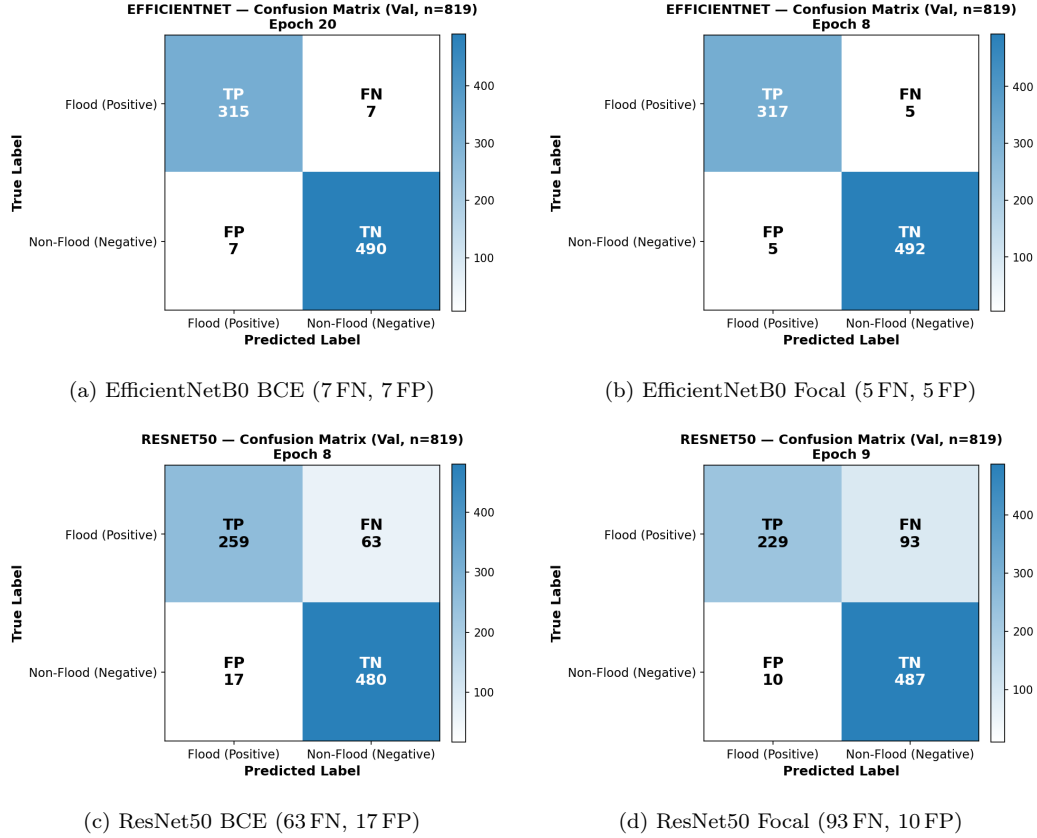


Figure 6: Confusion matrices for all four baseline conditions (val set, $n=819$, seed 42, $\tau=0.5$). EfficientNetB0 shows near-symmetric errors; ResNet50 shows systematic under-detection of floods. ResNet50 Focal (d) is operationally worse than ResNet50 BCE (c) despite having fewer FP, because 93 missed floods (28.9% miss rate) is unacceptable for a screening system.

5.3. *PR-AUC versus ROC-AUC: the metric matters*

Figure 7 illustrates the metric discrepancy directly. The ROC curves of both baseline BCE models appear respectable: ResNet50’s ROC-AUC of
515 0.9728 is only 2.6 points below EfficientNetB0’s 0.9985. A practitioner relying solely on ROC-AUC would conclude that ResNet50 is “nearly as good” as EfficientNetB0.

The precision-recall curves reveal the true picture. ResNet50’s precision collapses rapidly as recall exceeds approximately 0.85: to achieve 95 % flood
520 recall, a practitioner would need to accept a threshold at which precision drops substantially below the no-skill baseline. The PR-AUC gap—0.9976 vs. 0.9614, a difference of 3.6 points—directly quantifies this precision loss across the full recall range and accurately reflects the difference of 56 missed floods (63 FN vs. 7 FN).

The accuracy paradox compounds this picture. ResNet50 achieves 98.5 %
525 accuracy vs. EfficientNetB0’s 98.3 %—a difference of 0.2 points in the direction that would mislead a practitioner into preferring ResNet50. The mechanism is that ResNet50’s 63 missed floods reduce the accuracy score by $63/819 = 7.7\%$ (relative to the non-flood majority), but this is more than compensated by
530 ResNet50’s lower FP rate (17 vs. 7), which correctly classifies 10 additional non-flood images. PR-AUC correctly identifies EfficientNetB0 as the superior model for this recall-first screening problem. This empirically confirms the theoretical argument of Davis and Goadrich (2006) and the practical guidance of Saito and Rehmsmeier (2015).

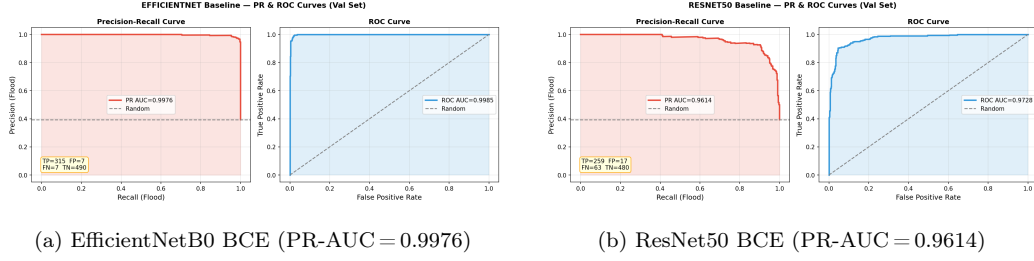


Figure 7: Precision-recall and ROC curves for the two BCE baselines (val set, seed 42). The 3.6-point PR-AUC gap (left panel vs. right panel) is substantially larger than the 2.6-point ROC-AUC gap, and directly reflects the difference of 56 missed floods. ResNet50’s precision collapses rapidly at high recall, while EfficientNetB0 maintains near-perfect precision across the full recall range.

5.4. Qualitative error analysis

Figure 8 shows the worst false positives (top rows, red border) and false negatives (bottom rows, orange border) for each BCE baseline at $\tau = 0.5$.

EfficientNetB0’s 7 false positives are river scenes photographed at close range: the images show reflective water surfaces, ripple patterns, and partially submerged vegetation or riverbank infrastructure that are visually indistinguishable from flooded road surfaces without wider field-of-view context. The 7 false negatives are flood scenes with attenuated water signal: dimly lit scenes, partially visible flooding in the background, or scenes where the dominant visual element is a dry foreground with flooding visible only in a portion of the frame. These are genuine boundary cases; a human reviewer presented with the same images would face similar uncertainty.

ResNet50’s error pattern is qualitatively different. While its false positives are also river-scene confounders, its 63 false negatives include many prototypical, unambiguous flood scenes—well-lit road surfaces covered with water,

550 visible debris, and clear flooding indicators that a human would immediately
 classify as flood. This pattern is characteristic of systematic decision-boundary
 collapse rather than ambiguity-driven misclassification: ResNet50’s boundary
 is miscalibrated, not merely challenged by hard cases. This distinction has
 important implications for remediation: the Phase-1 HNM approach is de-
 555 signed to address confounder-driven FP errors (relevant to EfficientNetB0)
 rather than systematic FN bias (the primary failure mode of ResNet50).

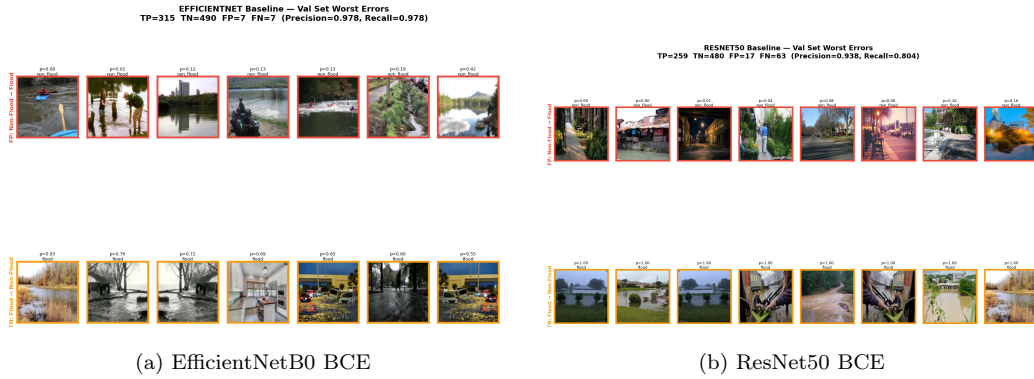


Figure 8: Worst false positives (red border, top rows) and false negatives (orange border, bottom rows) at $\tau=0.5$ (val set, seed 42). EfficientNetB0 FPs are visually ambiguous river scenes that share water texture with flood images. ResNet50 FNs include many prototypical flood scenes, indicating systematic rather than boundary-driven failure.

5.5. Confounder analysis

Table 4 reports per-category false positive rates on the validation set for both BCE baselines. This table is the primary deliverable of the confounder
 560 analysis: it shows which non-flood visual categories drive false positives at a
 statistically characterised level.

River images are the dominant confounder for both architectures. EfficientNetB0 BCE produces 7 river FPs in 76 validation river images (9.2%,

95 % CI: [3.8 %, 17.7 %]). ResNet50 BCE produces 3 river FPs (3.9 %, CI: 565 [0.8 %, 11.0 %]). The plant category produces 1 FP for EfficientNetB0 (2.9 %), zero for ResNet50. All other non-flood categories—animal, building, park walkway, street clear, swimming pool, vehicle—produce zero false positives at $\tau = 0.5$ for both models.

The interplay between the EfficientNetB0 and ResNet50 results is in- 570 structutive. EfficientNetB0 has a higher river FP rate (9.2 % vs. 3.9 %) but dramatically better flood recall (97.8 % vs. 80.4 %). The mechanism is a trade-off between flood sensitivity and non-flood specificity on the river category: EfficientNetB0’s more sensitive decision boundary catches more floods but also catches more rivers, whereas ResNet50’s less sensitive boundary misses 575 many floods while avoiding river FPs. For a recall-first screening system, the EfficientNetB0 operating point is strongly preferable: accepting 9.2 % river FP rate in exchange for 97.8 % flood recall versus 80.4 % flood recall is an advantageous trade.

The wide Clopper–Pearson CIs for categories with zero observed FP 580 deserve emphasis. For swimming pool ($N = 28$), the 95 % upper CI bound is 12.3 %—meaning that if a future deployment generated 28 new swimming pool images, we would expect with 95 % probability to see at least 0 and at most 3–4 FPs. A “zero FP on swimming pool” result does not imply robustness; it implies insufficient statistical power.

585 Figure 9 provides a heatmap view of the per-category false positive rates across all training conditions and architectures, complementing the confidence-interval analysis in Table 4.

Based on Table 4, the river category exceeds the flagging threshold $\theta = 0.05$

Table 4: Per-category false positive rates on the validation split, BCE baselines (seed 42, $\tau=0.5$). Clopper–Pearson 95 % binomial CIs via `scipy.stats.beta.interval`. Dagger ($\dagger$) marks water-confounder categories. Bold upper CI bounds flag categories where a zero observed FP rate should not be interpreted as robustness.

Category	N_{val}	EfficientNetB0 BCE		ResNet50 BCE	
		FP rate	95 % CI	FP rate	95 % CI
river $\dagger$	76	9.2 %	[3.8, 17.7]	3.9 %	[0.8, 11.0]
plant	34	2.9 %	[0.1, 15.3]	0.0 %	[0.0, 10.3]
swimming_pool $\dagger$	28	0.0 %	[0.0, 12.3]	0.0 %	[0.0, 12.3]
park_walkway $\dagger$	82	0.0 %	[0.0, 4.4]	0.0 %	[0.0, 4.4]
animal	67	0.0 %	[0.0, 5.4]	0.0 %	[0.0, 5.4]
building	55	0.0 %	[0.0, 6.5]	0.0 %	[0.0, 6.5]
street_clear	105	0.0 %	[0.0, 3.5]	0.0 %	[0.0, 3.5]
vehicle	50	0.0 %	[0.0, 7.1]	0.0 %	[0.0, 7.1]

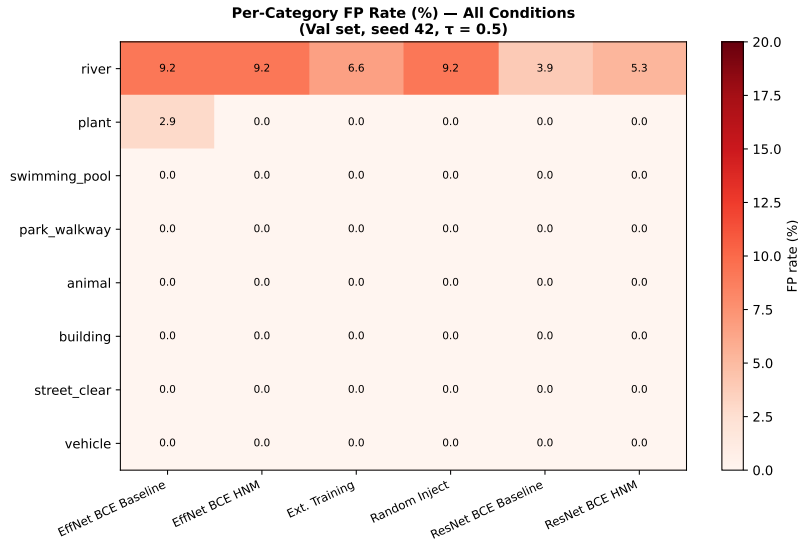


Figure 9: Per-category false positive rate heatmap across all training conditions and architectures (val set, $\tau=0.5$, seed 42). Colour intensity indicates FP rate; white cells denote zero observed FPs. River is the dominant confounder across all conditions. The heatmap complements Table 4 by showing condition-level variation at a glance.

for EfficientNetB0 (9.2 % > 5 %) and is at the boundary for ResNet50 (3.9 %
 590 < 5 %). Stage 1 of the P1-HNM pipeline therefore flags the river category for
 EfficientNetB0 and does not flag it for ResNet50. No other category exceeds
 $\theta = 0.05$ for either model.

5.6. Phase-1 HNM ablation results

Table 5 reports ablation results, Figure 11 shows the river false positive
 595 rates across all conditions with Clopper–Pearson confidence intervals, and
 Figure 10 compares PR and ROC curves for the four EfficientNetB0 ablation
 conditions.

Overall accuracy ordering. For EfficientNetB0 BCE, the strict or-
 dering HNM (98.78 %) > extended training (98.66 %) > random injection
 600 (98.41 %) > baseline (98.29 %) confirms the hierarchical isolation: (1) any
 additional training beyond Phase 2 improves performance over the baseline;
 (2) injecting category-matched data improves performance over extended
 training alone (random injection vs. extended training); (3) difficulty-ranked
 injection provides the largest gain (HNM vs. random injection). The 0.49
 605 accuracy percentage point improvement from HNM over baseline is the largest
 absolute accuracy gain across conditions (single seed; no significance test on
 overall accuracy has been conducted). The strict ordering is consistent with
 difficulty-ranked selection contributing beyond additional training epochs and
 category exposure alone.

610 For ResNet50 BCE, HNM improves accuracy by 0.13pp (98.66 % vs.
 98.53 %) over the baseline. The smaller gain is consistent with ResNet50’s
 fundamentally different failure mode: its 63 missed floods (FN bias) are not
 addressed by confounder-targeted HNM, which is designed to reduce FP rate

on the non-flood side of the boundary.

615 **River FP rates.** P1-HNM produced no statistically significant change
in river FP rate for either architecture (McNemar $p > 0.05$ after Bonfer-
roni correction for all pairwise comparisons; Clopper–Pearson CIs overlap
substantially across all conditions). For EfficientNetB0 BCE, P1-HNM and
baseline both produce 7/76 river FPs (9.2%). Extended training reduces this
620 to 5/76 (6.6%), and random injection returns to 7/76 (9.2%), but none of
these differences are detectable at $N_{\text{river}} = 76$. For ResNet50 BCE, P1-HNM
produces 4/76 (5.3%) versus baseline 3/76 (3.9%); this single-image change
is within noise. Power analysis indicates that a sample size of approximately
300 river validation images would be needed to detect a 3–4 percentage point
625 FP rate reduction at 80% power and $\alpha = 0.05$.

ResNet50 Focal HNM: decision-boundary collapse. The ResNet50
Focal HNM condition produced results consistent with the baseline focal
loss collapse: validation accuracy $\approx 87.4\%$, flood recall $\approx 71\%$, river FP
rate = 3.9% (same as baseline). The HNM pipeline cannot recover a model
630 whose decision boundary has collapsed under focal loss training. This finding
reinforces an important conclusion: gradient-level reweighting (focal loss) and
data-level augmentation (HNM) are not equivalent interventions and are not
interchangeable. For ResNet50, only BCE provides a viable training path;
HNM from a Focal baseline provides no benefit.

Table 5: P1-HNM ablation results (seed 42, $\tau=0.5$). Val Acc from best-epoch checkpoint. River FP rate from per-category confounder analysis. ΔAcc = change vs. respective baseline. Clopper–Pearson 95 % CI in brackets (%). McNemar pairwise tests on river FP status: all $p > 0.05$ after Bonferroni correction.

Model	Condition	Val Acc (%)	ΔAcc	River FP	River FP rate (%)
EfficientNetB0	Baseline	98.29	—	7/76	9.2 [3.8, 17.7]
EfficientNetB0	HNM (percentile)	98.78	+0.49 pp	7/76	9.2 [3.8, 17.7]
EfficientNetB0	Extended training	98.66	+0.37 pp	5/76	6.6 [2.2, 14.9]
EfficientNetB0	Random injection	98.41	+0.12 pp	7/76	9.2 [3.8, 17.7]
ResNet50	Baseline	98.53	—	3/76	3.9 [0.8, 11.0]
ResNet50	HNM (percentile)	98.66	+0.13 pp	4/76	5.3 [1.5, 13.0]

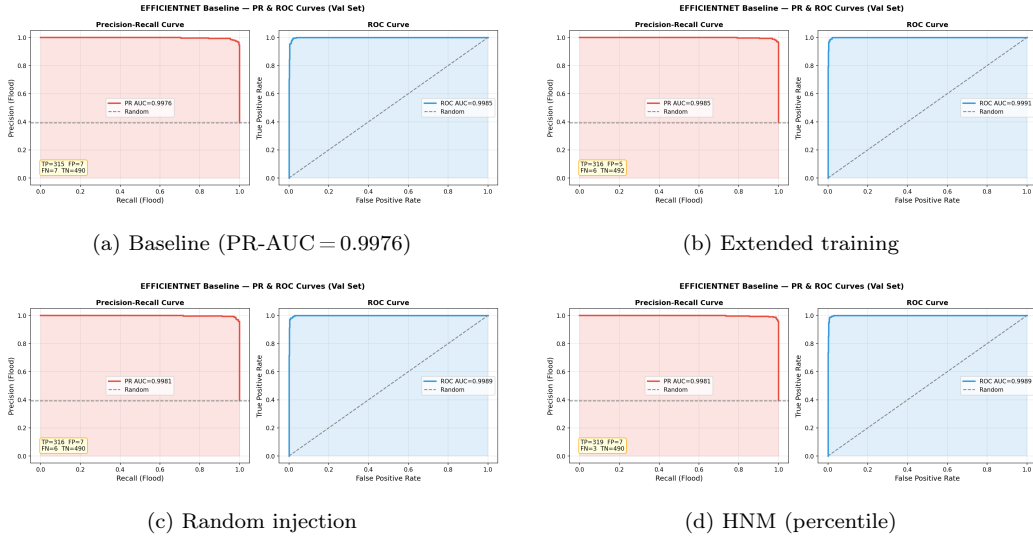


Figure 10: PR and ROC curves for the four EfficientNetB0 BCE ablation conditions (val set, seed 42). HNM maintains the highest PR-AUC, while the strict ordering HNM > extended training > random injection > baseline holds across the full recall range. Curves from the converged Phase-2 checkpoint for each condition.

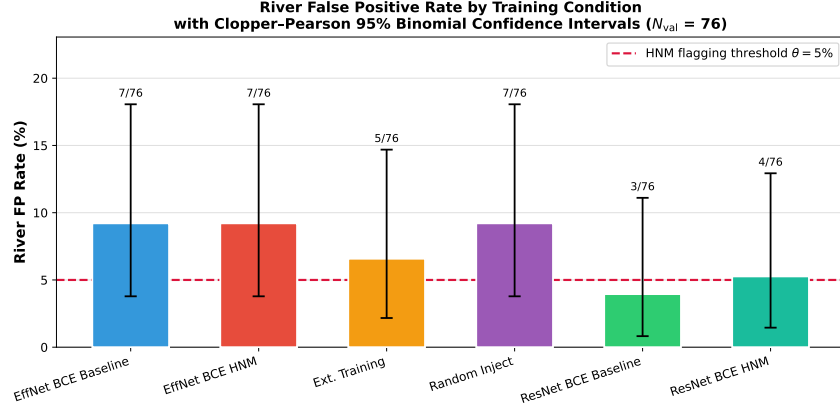


Figure 11: River false positive rate across all training conditions with Clopper–Pearson 95 % binomial confidence intervals ($N_{\text{river}}=76$ in all conditions). The dashed horizontal line marks the P1-HNM flagging threshold $\theta=5\%$. Wide CI overlap confirms that no pairwise difference is statistically significant at the available sample size. Annotated fractions show raw FP counts. Figure generated by `scripts/plot_confounder_fp.py`.

6. Discussion

6.1. Why PR-AUC reveals what accuracy hides

The accuracy paradox in Table 3 provides a concrete illustration of the metric selection problem. ResNet50 achieves *higher* overall accuracy (98.5 %) than EfficientNetB0 (98.3 %) yet misses nine times as many floods (63 vs. 7) and has PR-AUC that is 3.6 points lower. The mathematical mechanism is straightforward: the 819-image validation set contains 497 non-flood images (60.7 %) and 322 flood images (39.3 %). ResNet50’s systematic non-flood bias means it correctly classifies most of the 497 non-flood images (480/497) while misclassifying 63 of the 322 flood images. The accuracy calculation weights each correct classification equally, so the large non-flood majority dominates:

$$\frac{259+480}{819} = 90.2\% \text{ base accuracy without correctly classifying any floods would}$$

be 60.7 %. The 98.5 % figure reflects mostly non-flood accuracy with little flood accuracy contribution.

ROC-AUC reveals a meaningful difference (0.9985 vs. 0.9728) but quantifies it less sharply than PR-AUC because true negative rate (specificity) enters the denominator of the ROC calculation. With 480 true negatives out of 497 (96.6 %), the specificity is very high for both models, inflating ROC-AUC. PR-AUC replaces specificity with precision, directly penalising false positives that reduce precision; and it penalises false negatives through the recall axis. This double penalisation of both error types in the direction that matters for screening makes PR-AUC the appropriate primary metric.

We therefore recommend that all flood screening evaluations report PR-AUC as the primary metric alongside per-category FP rates with confidence intervals. Overall accuracy should be treated as a secondary metric at best; ROC-AUC as a supplementary one.

6.2. Architecture matters more than loss function for ResNet50

The divergent response to focal loss between EfficientNetB0 and ResNet50 is one of the more striking empirical findings of this paper. Both architectures were trained with balanced class weights, which provide global frequency correction for the flood:non-flood imbalance. Despite this, ResNet50 BCE achieves only 80.4 % flood recall, and focal loss ($\gamma = 2.0$) makes it substantially worse (71.1 %, 93 FN). EfficientNetB0 responds positively to both interventions: focal loss reduces FN from 7 to 5 and FP from 7 to 5.

The explanation likely lies in the quality of the learned flood representation. EfficientNetB0’s compound-scaled architecture (Tan and Le, 2019) balances depth, width, and resolution, producing richer feature representations from

ImageNet transfer that encode fine-grained texture distinctions relevant to flood detection. Near the decision boundary, EfficientNetB0’s representations are diverse enough that gradient reweighting (focal loss) can find a better
675 boundary by focusing on the most informative hard cases. ResNet50, with its larger parameter count (25.6 M vs. 5.3 M), appears to overfit the dominant non-flood pattern despite balanced class weighting, producing a decision boundary that systematically assigns low flood probability. In this regime, focal loss amplifies the problem: by down-weighting the well-classified easy
680 non-flood examples, it further reduces gradient signal from the misclassified flood examples, accelerating the collapse.

The practical implication is important: focal loss is not a universally beneficial intervention for flood screening, and its effect depends on whether the architecture’s feature space has learned a quality flood representation.
685 Class weighting ($\neq$ focal loss) is the safer default; focal loss should be applied only after verifying that the model’s flood recall is already near-plateau.

6.3. Why Phase-1 mining works: convergence, uncertainty, and the offline HNM constraint

As described in Section 4.4, training convergence eliminates the mining sig-
690 nal; the Phase-1 checkpoint retains genuine uncertainty on visual confounders that the Phase-2 checkpoint has resolved.

Phase-1 mining solves this by identifying an intermediate checkpoint at which the model retains meaningful uncertainty on hard negatives. This is analogous to knowledge distillation (Guo et al., 2017): intermediate represen-
695 tations in partially trained models often encode more generalisable information than the final model, which has “compressed” its uncertainty into confident

predictions. By using the Phase-1 model’s uncertainty as a proxy for the visual ambiguity between flood and confounder categories, we recover a useful signal that the Phase-2 model has suppressed.

700 The improvement in overall classification accuracy (0.49 accuracy percentage points for EfficientNetB0) provides empirical evidence that Phase-1 mining identifies images that are genuinely informative for refining the decision boundary, rather than simply adding more training data. The strict ordering HNM > extended training > random injection isolates this contribution to
705 the difficulty ranking itself.

6.4. Interpreting the HNM river FP results

The null result for river FP rate reduction under P1-HNM merits careful interpretation. The HNM pipeline produces 7/76 river FPs both before and after mining for EfficientNetB0—identical to baseline—while overall
710 accuracy improves by 0.49 accuracy percentage points. Several explanations are consistent with this pattern.

First, the improvements in overall accuracy may arise from better classification of flood images at the decision boundary (reduced FN) rather than from better rejection of river images (reduced FP). The additional hard negatives
715 from the HNM pipeline correct the model’s boundary in the flood-adjacent region, improving flood recall, while the river-adjacent region of the boundary—corresponding to the most visually ambiguous river images—remains difficult regardless of training condition.

Second, the validation sample size is insufficient to detect the difference.
720 With $N_{\text{river}} = 76$ and an observed FP rate near 9%, a reduction of 2 FPs (from 7 to 5) would be needed to reach the extended-training level of 6.6%;

this requires a sample of approximately 300 river images to be detectable at standard power levels. The null result is consistent with no true effect, with a true effect below the detectable threshold at this sample size, or with genuine
725 visual indistinguishability of the hardest river images from floods.

Third, the most visually confusing river images—the 7 FPs in both baseline and HNM conditions—may be genuinely indistinguishable from floods without additional context, in which case no training intervention short of providing additional contextual cues would reduce the error.

730 6.5. Limitations

- **Single seed.** All results are from seed = 42; multi-seed replication (seeds 1, 7, 123, 2024) is in progress. Bootstrap CIs and McNemar tests cannot be reported until seed aggregation is complete. This limits the statistical strength of the ablation conclusions.
- 735 • **Small non-flood N for water categories.** Swimming pool ($N_{\text{val}}=28$) and plant ($N_{\text{val}}=34$) have validation sets too small to draw conclusions about confounder robustness. The very categories most operationally relevant for false-positive reduction (pools are common near flooded residential areas) are the categories where we have the least statistical
740 power.
- **Single geographic and source distribution.** The dataset sources have not been fully characterised for temporal or geographic clustering; USF FloodingDataset images are primarily from North American flood events. The RIWA river images come from diverse European and North

745 American river systems. Generalisation to tropical, Asian, or African
flood environments is not validated.

- **Phase boundary not ablated.** The (30, 50) freeze boundary is a fixed design choice motivated by prior progressive fine-tuning literature; optimal values for flood-specific data are unknown. Different boundary
750 choices might produce stronger or weaker Phase-1 mining signals.
- **No external test set.** All evaluation is on the held-out validation split. Out-of-distribution generalisation to images from different cameras, weather conditions, flood types, or geographic locations is not assessed. External validation should be a priority for any deployment.
- 755 • **Phase-2 mining produces zero candidates.** By the end of Phase 2, the fully converged model assigns near-zero flood probability to all training non-flood images. Phase-2 mining was therefore not pursued; this structural property of converged offline models is the direct motivation for Phase-1 mining. Offline HNM from fully converged checkpoints is
760 a known limitation reported in other applications (Viola and Jones, 2004).

6.6. Practitioner recommendations

Based on the evidence in this paper, we offer the following practical recommendations for teams building street-level flood screening systems:

- 765 1. **Report PR-AUC** as the primary evaluation metric, not accuracy or ROC-AUC. Report per-category FP rates with Clopper–Pearson confidence intervals for any category with $N_{\text{val}} < 100$.

2. **Prefer EfficientNetB0** over ResNet50 for binary flood screening.
EfficientNet’s compound-scaled features provide substantially better
770 flood recall (97.8 % vs. 80.4 %), the primary operational requirement
for a screening system.
3. **Apply P1-HNM** when the Phase-1 river or pool FP rate exceeds 5 %.
Mine from the Phase-1 checkpoint, not the converged Phase-2 model.
Expect improvement in overall accuracy; expect that FP rate reduction
775 will require larger validation sets to confirm statistically.
4. **Do not apply focal loss** to models that already exhibit systematic
flood under-detection. Verify that flood recall is near-plateau before
applying focal loss; use BCE + balanced class weights as the baseline.
5. **Expand water-confounder validation sets** before deployment, par-
780 ticularly for swimming pools and lakes which may co-occur spatially
with floods in residential areas.

7. Conclusion

We presented Phase-1 Hard Negative Mining (P1-HNM), a three-stage
pipeline for improving street-level flood detection by exploiting an intermedi-
785 ate training checkpoint as a mining oracle. The core insight is that training
convergence destroys the mining signal needed for productive hard negative
selection: a fully converged Phase-2 model assigns near-zero flood probability
to all training non-flood images, leaving no candidates. The Phase-1 check-
point, which has trained only the classification head and top backbone layers,
790 retains genuine uncertainty on visually ambiguous categories and provides a
meaningful difficulty ranking.

On a 4,099-image benchmark spanning 11 fine-grained categories, we demonstrated that EfficientNetB0 achieves PR-AUC = 0.9976 versus ResNet50’s 0.9614—a 3.6-point gap that is invisible in ROC-AUC (0.9985 vs. 0.9728).
 795 This empirically confirms that PR-AUC is the appropriate primary metric for this class-imbalanced screening problem. The gap corresponds to a difference of 56 missed floods (63 vs. 7) in an 819-image validation set, a difference that aggregate accuracy (98.5 % vs. 98.3 %) actually inverts.

Phase-1 HNM improves overall classification accuracy by 0.49 accuracy
 800 percentage points for EfficientNetB0 BCE, outperforming extended training (+0.37 pp) and random injection (+0.12 pp), confirming that difficulty-ranked selection contributes beyond additional training epochs and category-exposure alone. The per-category river FP reduction does not reach statistical significance at $N_{\text{river}} = 76$; power analysis indicates that approximately 300 river
 805 validation images would be needed to detect a 3–4 percentage point reduction.

Per-category FP analysis with Clopper–Pearson confidence intervals reveals that river is the dominant confounder for both architectures; all other categories produce zero false positives at $\tau = 0.5$. Wide CIs for small- N categories (swimming pool, plant) highlight that “zero FP” cannot be equated
 810 with confounder robustness without larger evaluation sets.

We recommend that flood screening systems be evaluated at the per-category FP rate level with explicit confidence intervals, that PR-AUC replace accuracy and ROC-AUC as the primary reported metric, and that Phase-1 mining be applied when any water-confounder category produces a training-set
 815 FP rate above 5 %. Future work should expand category-specific validation samples, conduct multi-seed replication to provide bootstrap confidence inter-

vals, and evaluate out-of-distribution generalisation on geographically diverse datasets.

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820 [ANONYMISED FOR REVIEW]

Data and code availability

Training dataset: `zinnia82/flood-binary-hnm-benchmark` (Hugging Face Hub). River augmentation images: RIWA dataset (Wagner et al., 2023), <https://www.kaggle.com/datasets/franzwagner/river-water-segmentation-dataset>.

825 Source code: [TODO — GitHub link before submission].

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Appendix A. Hyperparameters

Table A.6 lists all training hyperparameters. Phase-1 and Phase-2 learning rate values are verified against the source code constants `PHASE1_LR = 1e-4`

905 and PHASE2_LR=1e-5 in scripts/train_baseline.py, correcting an earlier
draft that incorrectly listed 10^{-3} and 10^{-4} .

Appendix B. Clopper–Pearson CI reference

Table B.7 reports the upper bound of the 95 % Clopper–Pearson CI for
an observed FP rate of 0 % ($k=0$) as a function of validation sample size.
910 This reference table helps readers interpret the “0 % FP” results in Table 4: a
zero observed rate does not mean zero true rate, and the uncertainty grows
substantially as N decreases.

Table A.6: Training hyperparameters verified against source code.

Hyperparameter	Value
Phase-1 learning rate	1×10^{-4}
Phase-2 learning rate	1×10^{-5}
Phase-1 max epochs	15
Phase-2 max epochs	20
Early stopping patience	7 epochs (val loss)
LR reduce patience (Phase 1)	3 epochs
LR reduce patience (Phase 2)	4 epochs
Batch size	32
Input image size	224×224
Dropout rates	0.2 (pre-Dense), 0.3 (post-BN)
Dense hidden units	256
Focal loss γ	2.0
Phase-1 trainable layers (top n)	30
Phase-2 frozen layers (first n)	50
HNМ mining checkpoint	Phase-1 best val loss
HNМ retraining start	Phase-2 checkpoint
HNМ top percentile	10 %
HNМ FP flagging threshold θ	0.05
HNМ augmentation factor	$5\times$ per mined image
HNМ augmentation operations	Rotation $\pm 30^\circ$, zoom $\pm 30\%$, brightness $[0.6, 1.4]$, H-flip
Decision threshold τ	0.5
Random seed	42 (single seed; multi-seed pending)
Compute platform	Apple M4 (16 GB, Metal GPU)

Table B.7: Upper bound of 95 % Clopper–Pearson CI for $k=0$ FP observed, by sample size n . Categories in the present study are listed in parentheses. The upper CI bound for swimming pool (12.3 %) implies that a true FP rate of up to 12.3 % is consistent with the observed zero-FP result.

n (val images)	Upper 95 % CI bound
10	30.8 %
15	21.8 %
28 12.3 % (swimming pool)	
34 10.3 % (plant)	
50 7.1 % (vehicle)	
55 6.5 % (building)	
67 5.4 % (animal)	
76 4.7 % (river)	
82 4.4 % (park_walkway)	
105 3.5 % (street_clear)	